

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		[For a body in equilibrium the sum of] the clockwise <u>moments</u> = [the sum of] the anticlockwise <u>moments</u> [about the same point]	1			1		1
		(ii)		Substitution: moment = 2×40 (1) Moment = 80 [N cm] (1)	1	1		2	2	2
		(iii)		Distance = $\frac{80(\text{ecf})}{5}$ (1) Distance = 16 [cm] (1)	1	1		2	2	2
	(b)			Moment will <u>double</u> (1) so the distance [from the pivot] needs to <u>double</u> so she is wrong (1) Alternative $4 \times 40 = 160$ [N cm] (1) so the distance [from the pivot] needs to <u>double or is 32 cm</u> so she is wrong (1) To award full marks the correct conclusion needs to be present			2	2		2
				Question 4 total	3	2	2	7	4	7